

Self-Introduction & Research Plan at HCMUT

Eita Matsuura

GD1, Science of Technology Innovation (Year 1 of 5-year Integrated Doctoral Program)

Theory of Machine Learning Laboratory (TML Lab)
Nagaoka University of Technology

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About Me

Eita Matsuura (松浦 瑛太)

Program:

- 5-year Integrated Doctoral Program, Science of Technology Innovation (GD1, equivalent to M1)

University:

- Nagaoka University of Technology

Lab:

- Theory of Machine Learning Laboratory

Supervisor:

- Assoc. Prof. Gendo Kumoi

Research Interest

- Effectiveness evaluation of a generative-AI career-education support system for high school students

Hobbies

- Finding good restaurants
- Reading manga

Background: Existing Research

Two known phenomena motivating my research

1. LLM low-diversity problem

- LLMs produce low-variance responses by default
- Cannot reproduce the diversity of real human decision-making

2. Foreign Language Effect (Keysar et al., 2012)

- People's decision-making changes between their L1 and L2
- A well-documented cognitive bias in human studies

Open question: Does the Foreign Language Effect propagate into LLM-based simulation of human decision-making?

Research Topic

Evaluating LLM Simulation Diversity with Real Decision-Making Styles under Japanese–Vietnamese × L1/L2 Cross-Cultural Conditions

Methodology: Culture × Language 2×2 Design

	Style doc from JP participants	Style doc from VN participants
Interview in L1	① JP × Japanese (L1)	② VN × Vietnamese (L1)
Interview in L2	③ JP × English (L2)	④ VN × English (L2)

- Per condition: **3 tasks × 3 LLMs** full factorial experiment
- **Evaluation metric:** Multilingual Vendi Score (diversity)
- **Participants:** N=15 Japanese (NUT) + N=15 Vietnamese (HCMUT)

Research Plan

3-Phase Research Plan

Phase	Period	Location	Key Activities
Phase 1	– Aug 2026	Japan (NUT)	JP data (N=15, L1/L2 pairs); build multilingual Vendi Score framework
Phase 2	Sep – Nov 2026	Vietnam (HCMUT)	VN data (N=15, VN/EN pairs); JP × VN × L1/L2 × 3 tasks × 3 LLMs experiment
Phase 3	Dec 2026 –	Japan (NUT)	4-condition cross-analysis; culture/language invariant extraction; paper writing

Expected outputs:

- JP–VN × L1/L2 paired style-document corpus (N=30, open on GitHub)
- Multilingual Vendi Score framework (OSS)
- International co-authored paper

Collaboration Request

What We Need from HCMUT

1. **Participant Recruitment** — ~N=15 Vietnamese participants
 - Style documents collected via AI interviews in Vietnamese (L1) + English (L2)
2. **Your Support** — We warmly welcome any collaboration and support from Prof. Tho's laboratory in making this research a success.

Thank You for Listening!

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